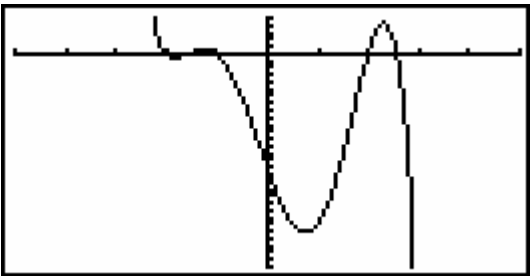
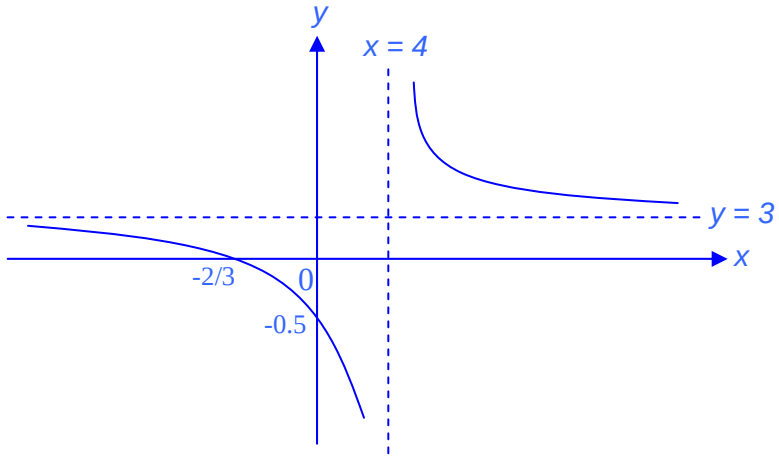
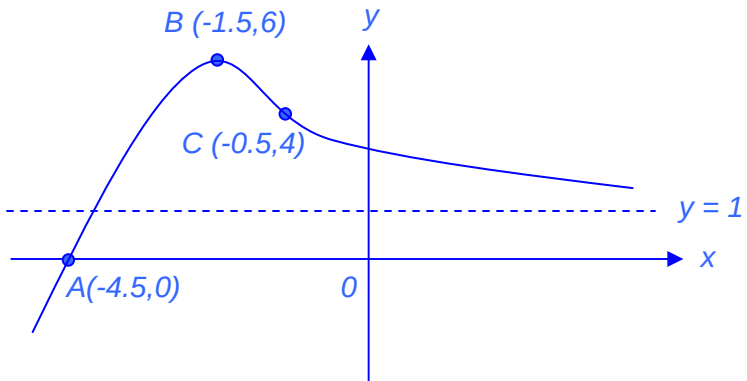
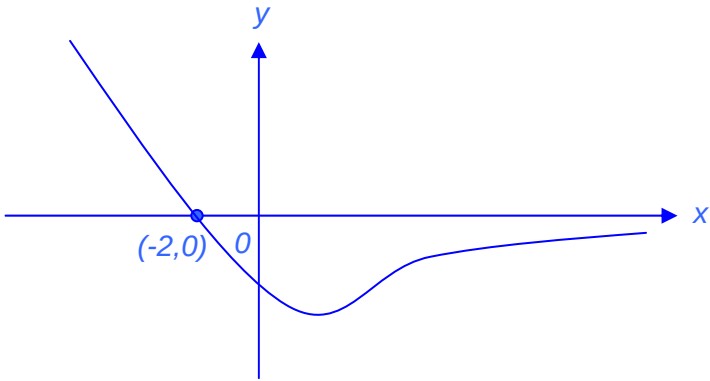


1		$(x+4)(x^2-4) \geq x(x^2-4)^2$ $(x+4)(x^2-4) - x(x^2-4)^2 \geq 0$ 
		$x < -2,$ $-1.56 \leq x \leq -1,$ $2 < x \leq 2.56$
2	(i)	$\frac{2}{n(n-1)} = \frac{A}{n} + \frac{B}{n-1}$ $A = \frac{2}{-1} = -2$ $B = \frac{2}{1} = 2$ $\frac{2}{n(n-1)} = -\frac{2}{n} + \frac{2}{n-1}$
	(ii)	$\sum_{r=2}^N \frac{1}{r(r-1)}$ $= \left[-\frac{1}{2} + \frac{1}{1} \right]$ $+ \left[-\frac{1}{3} + \frac{1}{2} \right]$ $+ \left[-\frac{1}{4} + \frac{1}{3} \right]$ $+ \dots$ $+ \left[-\frac{1}{N} + \frac{1}{N-1} \right]$ $= 1 - \frac{1}{N}$ $\text{As } N \rightarrow \infty, \sum_{r=2}^{\infty} \frac{1}{r(r-1)} = 1.$

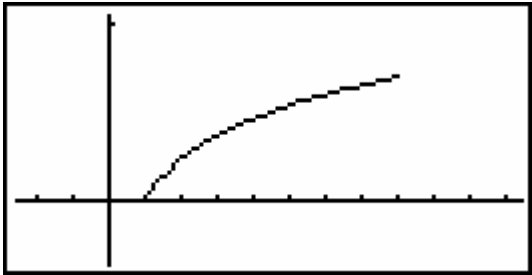
3	$D + M + W = 800 \dots\dots(1)$ $5.2D + 3.1M + 2.5W = 2540 \dots\dots(2)$ $4.8D + 2.9M + 1.50W = 2160 \dots\dots(3)$ $\begin{pmatrix} 1 & 1 & 1 \\ 5.2 & 3.1 & 2.5 \\ 4.8 & 2.9 & 1.5 \end{pmatrix} \begin{pmatrix} D \\ M \\ W \end{pmatrix} = \begin{pmatrix} 800 \\ 2540 \\ 2160 \end{pmatrix}$ $\begin{pmatrix} D \\ M \\ W \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 5.2 & 3.1 & 2.5 \\ 4.8 & 2.9 & 1.5 \end{pmatrix}^{-1} \begin{pmatrix} 800 \\ 2540 \\ 2160 \end{pmatrix}$ $\begin{pmatrix} D \\ M \\ W \end{pmatrix} = \begin{pmatrix} 100 \\ 450 \\ 250 \end{pmatrix}$ The factory should stop producing White chocolate bars as it has the least profit margin.
4	(i) Let G_n represent the nth term of the GP. $G_1 = a$ $G_2 = a + 4d$ $G_3 = a + 6d$ $G_2 / G_1 = (a + 4d) / a$ $r = 1 + 4(d/a)$ $d/a = (r - 1) / 4$ $G_3 / G_1 = (a + 6d) / a$ $r^2 = 1 + 6(d/a)$ $d/a = (r^2 - 1) / 6$ $(r - 1) / 4 = (r^2 - 1) / 6$ $6(r - 1) = 4(r^2 - 1)$ $6r - 6 = 4r^2 - 4$ $4r^2 - 6r + 2 = 0$ $2r^2 - 3r + 1 = 0$ (ii) $2r^2 - 3r + 1 = 0$ $(2r-1)(r-1) = 0$ $r = 0.5$ $r = 1$ (NA) Since $r < 1$, the geometric series is convergent. (iii) $S_n = [a (1 - r^n)] / (1 - r)$ $S_n = [a (1 - 0.5^n)] / (1 - 0.5)$ $S_n = 2a (1 - 0.5^n)$

5(a)	(i)	By long division, $y = 3$, $x = 4$
	(ii)	
	(iii)	$kx = \frac{3x + 2}{x - 4}$
		$kx^2 - 4kx - 3x - 2 = 0$ $kx^2 + (-4k - 3)x + (-2) = 0$ $(-4k - 3)^2 - 4(k)(-2) > 0$ $\dots$ $k < -1.66 \text{ or } k > -0.339 \text{ (3 s.f.)}$

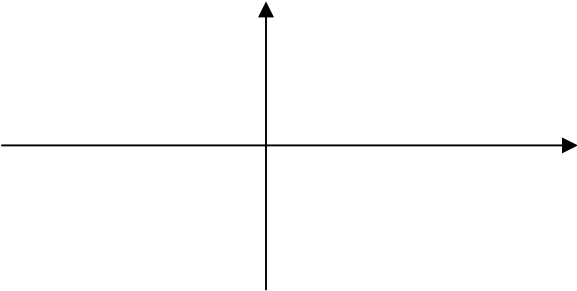
5(b)	(i)	
5(b)	(ii)	
6(a)		$\begin{aligned} \frac{\sin 2x}{\sqrt{\cos x}} &\approx \left[2x - \frac{(2x)^3}{3!} + \dots \right] \left(1 - \frac{x^2}{2} \right)^{-\frac{1}{2}} \\ &= \left[2x - \frac{(2x)^3}{3!} + \dots \right] \left[1 + \left(-\frac{1}{2} \right) \left(-\frac{x^2}{2} \right) + \dots \right] \\ &= \left(2x - \frac{4}{3}x^3 + \dots \right) \left[1 + \frac{x^2}{4} + \dots \right] \\ &= 2x - \frac{5x^3}{6} + \dots \end{aligned}$

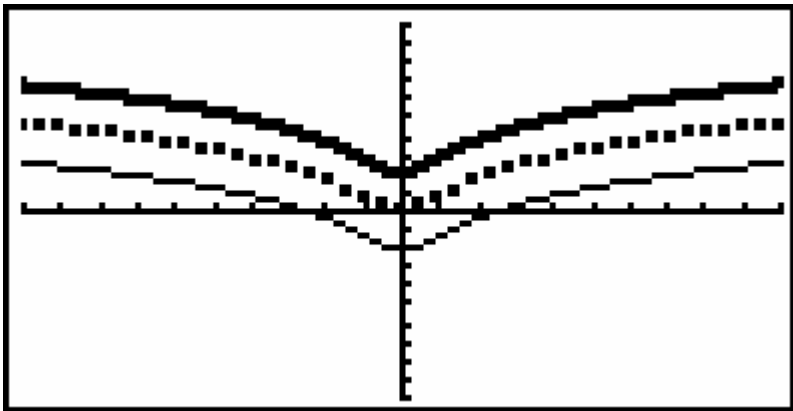
	(i) By similar triangles, $\frac{r}{3} = \frac{6-h}{6}$ $r = 3 - \frac{1}{2}h$ Hence, $V = \frac{1}{3}\pi(3 - \frac{1}{2}h)^2(6-h)$. $V = \frac{1}{3}\pi(3 - \frac{1}{2}h)^2(6-h)$ $\frac{dV}{dh} = \frac{1}{3}\pi\left[(2)(3 - \frac{1}{2}h)(6-h)(-\frac{1}{2}) + (3 - \frac{1}{2}h)^2(-1)\right]$ $\frac{dV}{dh} = -\frac{1}{3}\pi(3 - \frac{1}{2}h)(6-h) - \frac{1}{3}\pi(3 - \frac{1}{2}h)^2$ When $h = 1$, $\frac{dV}{dh} = \frac{1}{3}\pi\left[(2)(3 - \frac{1}{2}(1))(6-(1))(-\frac{1}{2}) + (3 - \frac{1}{2}(1))^2(-1)\right]$ $\frac{dV}{dh} = -\frac{25}{4}\pi$ $\frac{dh}{dt} = -\frac{4}{25}\pi \text{ cms}^{-1}$
7	(i) $l:r = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1-1 \\ 4-2 \\ 7-3 \end{pmatrix}$ $l:r = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ $\sin \theta = \frac{\left \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right }{\sqrt{3^2 + (-4)^2 + 1^2} \cdot \sqrt{0^2 + 1^2 + 2^2}}$ $\theta = \sin^{-1}\left(\frac{0-4+2}{\sqrt{26} \cdot \sqrt{5}}\right)$ $\theta = 0.176 \text{ rad or } 10.1^\circ$

	(ii)	Let the l_1 represent the line containing point A, perpendicular to the plane. Let F represent the foot of the perpendicular. $l_2 : r = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ $\left[\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} \right] \cdot \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} = 11$ $3 + 9\alpha - 8 + 16\alpha + 3 + \alpha = 11$ $26\alpha = 13$ $\alpha = \frac{1}{2}$ $\overrightarrow{OF} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ $\overrightarrow{OF} = \begin{pmatrix} 2\frac{1}{2} \\ 0 \\ 3\frac{1}{2} \end{pmatrix}$
	(iii)	$RREF \begin{pmatrix} 3 & -4 & 1 & 11 \\ 1 & 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & \frac{5}{7} & \frac{15}{7} \\ 0 & 1 & \frac{2}{7} & -\frac{8}{7} \end{pmatrix}$ $x = \frac{15}{7} - \frac{5}{7}z$ $x + \frac{5}{7}z = \frac{15}{7} \quad y = -\frac{8}{7} - \frac{2}{7}z$ $y + \frac{2}{7}z = -\frac{8}{7} \rightarrow z = z$ $r = \begin{pmatrix} \frac{15}{7} \\ -\frac{8}{7} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} \frac{5}{7} \\ \frac{2}{7} \\ -1 \end{pmatrix}$ Line of intersection l:
8	(i)	$\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = \frac{1}{t}$

		$\frac{dy}{dx} = \frac{1}{3t^3}$ $\frac{dy}{dx} = \frac{1}{3x}$
	(ii)	 (10) and (8, ln 2)
	(iii)	When $t = \frac{3}{2}$, $x = \frac{27}{8}$, $y = \ln\left(\frac{3}{2}\right)$, $m = \frac{1}{3\left(\frac{3}{2}\right)^3}$ $m = \frac{8}{81}$ $y = \frac{8}{81}x + c$ $c = \ln\left(\frac{3}{2}\right) - \frac{1}{3}$ $y = \frac{8}{81}x + \ln\left(\frac{3}{2}\right) - \frac{1}{3}$ $y = \ln\left(\frac{3}{2}\right) - \frac{1}{3}$ $x = \frac{81}{8} \left[\frac{1}{3} - \ln\left(\frac{3}{2}\right) \right]$ Area of triangle AOB $= \frac{1}{2} \left[\left[\ln\left(\frac{3}{2}\right) - \frac{1}{3} \right] \frac{81}{8} \left[\frac{1}{3} - \ln\left(\frac{3}{2}\right) \right] \right]$ $= 0.0263 \text{ units}^2 \text{ (3sf)}$

9	(i)	$u_2 = \frac{1}{2},$ $u_3 = \frac{1}{3},$ $u_4 = \frac{1}{4}$
	(ii)	$u_n = \frac{1}{n}$
	(iii)	Let P_n be the statement $u_n = \frac{1}{n}$. P_1: when $n = 1$ $\frac{1}{1} = 1$ $u_1 = 1$ P_1 is true. Assume P_k is true for some $k \geq 1$. $u_k = \frac{1}{k}$ P_{k+1}: when $n = k+1$ $u_{k+1} = \frac{u_k}{u_k + 1}$ $= \frac{\frac{1}{k}}{\frac{1}{k} + 1}$ $= \frac{\frac{1}{k}}{\frac{1+k}{k}}$ $= \frac{1}{k+1}$ P_{k+1} is true. Since P_1 is true and P_k is true implies P_{k+1} is true, by Mathematical Induction.

10(a))	(i)	$z^4 = 16e^{i\left(-\frac{\pi}{2}+2k\pi\right)}$ $z = 2e^{i\left(\frac{-\frac{\pi}{2}+2k\pi}{4}\right)}, k = -1, 0, 1, 2$
	(ii)	
10(b))		$(1+2i)^3 + a(1+2i)^2 + b(1+2i) + 2 = 0$ $-9-2i + a(-3+4i) + b(1+2i) = 0$ $b = \frac{9+2i-a(-3+4i)}{1+2i} = 2.6-3.2i-a(1+2i)$ $(1-2i)^3 + a(1-2i)^2 + b(1-2i) + 2 = 0$ $-9+2i + a(-3-4i) + b(1-2i) = 0$ $-9+2i + a(-3-4i) + [2.6-3.2i-a(1+2i)](1-2i) = 0$ $-12.8-6.4i + a(-8-4i) = 0$ $a = -1.6 \quad b = 4.2$
11(a))	(i)	$\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$ $= \int_0^{\frac{\pi}{2}} \frac{1}{2}(1-\cos 2x) \, dx$ $= \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} 1 \, dx - \int_0^{\frac{\pi}{2}} \cos 2x \, dx \right]$ $= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{\sin \pi}{2} \right) - \left(0 - \frac{\sin 0}{2} \right) \right]$ $= \frac{\pi}{4}$

	(ii)	$\int e^x \sin x \, dx = e^x(-\cos x) - \int (-\cos x)e^x \, dx$ $\int e^x \sin x \, dx = -e^x \cos x + \left[e^x \sin x - \int (\sin x)e^x \, dx \right]$ $2 \int e^x \sin x \, dx = -e^x \cos x + e^x \sin x + d$ $\int e^x \sin x \, dx = \frac{1}{2} e^x (\sin x - \cos x) + c$
11(b)	(i)	$\left(\frac{x^2 + 1}{x} \right) \frac{dy}{dx} = 2$ $\frac{dy}{dx} = \frac{2x}{x^2 + 1}$ $y = \ln(x^2 + 1) + c$
	(ii)	When $x = 0$ and $y = 0$, $c = 0$. Particular solution: $y = \ln(x^2 + 1)$
	(iii)	
	(iv)	Gradient tends towards zero as x increases to become extremely large.